

Program: Diploma in Artificial Intelligence and Machine Learning	
Course Code: 6341B	Course Title: Robotic Process Automation
Semester: 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L: 4 T: 0 P: 0)	Periods per semester: 60

Course Objectives:

- To understand the basic concepts of automation.
- To learn about different types of bots.
- To learn and understand Robotic Process Automation and their working.
- To learn about different RPA tools.
- To understand the challenges & risks when implementing RPA.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic programming concepts	2131	Problem Solving and Programming	2

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the concepts of RPA and its architecture.	13	Understand
CO2	Explain the robotic control flow architecture and tools used in RPA.	17	Understand

CO3	Outline RPA life cycle and its implementation	14	Understand
CO4	Describe the working of RPA with existing systems and the features of various RPA tools.	14	Understand
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the concepts of RPA and its architecture.		
M1.01	Explain automation, scope and techniques	2	Understanding
M1.02	Illustrate the concepts and benefits of RPA	2	Understanding
M1.03	Explain RPA services	2	Understanding
M1.04	Explain bots and types of bots	2	Understanding
M1.05	Describe the architecture of RPA	2	Understanding
M1.06	Explain programming elements in RPA	3	Understanding

Contents:
Fundamentals of RPA: - What is automation? Scope of automation, techniques of automation, processes and flow charts. RPA - benefits of RPA, Automation vs RPA, RPA services, advantages and disadvantages. Bots-types of bots.
RPA Architecture – Applications under Robotic Process Execution, RPA tools, RPA platform, RPA execution infrastructure.
Programming elements in RPA – Recorder, Development studio, Plug-in/Extension, Bot runner, control center.

CO2:	Explain the robotic control flow architecture and tools used in RPA.		
M2.01	Explain RPA Development methodologies	2	Understanding
M2.02	Illustrate the Robotic control flow architecture with examples	3	Understanding
M2.03	Explain Process Design Document/Solution Design Document	2	Understanding.
M2.04	Explain the different types of variables and attributes with examples	3	Understanding
M2.05	Describe the control flow activities with examples	4	Understanding.
M2.06	Explain basic, desktop and web recording	2	Understanding.
M2.07	Describe screen scraping with an example	1	Understanding.
	Series test -1	1	

Contents:
Standardization of processes - RPA Development methodologies -Robotic control flow architecture - RPA business case - Process Design Document/Solution Design Document - Industries best suited for RPA - Risks & Challenges with RPA.
Introduction to UiPath –Variables (Managing Variables, Naming Best Practices, The Variables Panel, Generic Value Variables, Text Variables, True or False Variables, Number Variables, Array Variables, Date and Time Variables, Data Table Variables), Control Flow (If Else Statements, Loops, Advanced Control Flow, Sequences, Flowchart), Control Flow Activities (The Assign Activity, The Delay Activity, The Do While Activity, The If Activity, The Switch Activity, The While Activity, The For Each Activity, The Break Activity), Arguments, Recording -Recording Introduction, Basic and Desktop Recording, Web Recording, Input/Output Methods. Screen scraping

CO3	Outline RPA life cycle and its implementation		
M3.01	Explain the life cycle of RPA-discovery phase	3	Understanding
M3.02	Describe solution design phase and development phase	3	Understanding
M3.03	Describe user acceptance test (UAT) and deployment and maintenance phase.	2	Understanding.
M3.04	Explain execute bots phase.	3	Understanding.
M3.05	Describe RPA implementation.	3	Understanding
Contents: RPA life cycle – Discovery phase, solution design phase, development phase, user acceptance test (UAT), deployment and maintenance phase, execute bots. RPA implementation.			
CO4:	Describe the working of RPA with existing systems and the features of various RPA tools.		
M4.01	Explain the working of RPA	2	Understanding
M4.02	Explain the working of AI powered RPA	2	Understanding
M4.03	.Describe RPA tools and its features	5	Understanding
M4.04	Describe the core functionalities of RPA tools	3	Understanding
M4.05	Describe the parameters for choosing the right RPA tool.	2	Understanding
	Series Test - 2	1	
Contents: Working of RPA-with the existing systems,-back end and front end integrations, working of AI powered RPA. RPA tools-UiPath, Blue Prism, Automation Anywhere, Pega, WorkFusion,- Core Functionalities of RPA tools. Parameters for choosing the right RPA tool.			

Text / Reference

T/R	Book Title/Author
T1	Alok Mani Tripathi, “Learning Robotic Process Automation” , Packt Publishing, 2018
R1	Frank Casale , Rebecca Dilla, Heidi Jaynes , Lauren Livingston, “Introduction to Robotic Process Automation: a Primer” , Institute of Robotic Process Automation, 1st Edition 2015.
R2	Richard Murdoch, Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant” , Independently Published, 1st Edition 2018.
R3	Srikanth Merianda, “Robotic Process Automation Tools, Process Automation and their benefits: Understanding RPA and Intelligent Automation” , Consulting Opportunity Holdings LLC, 1st Edition 2018.
R4	Lim Mei Ying, “Robotic Process Automation with Blue Prism Quick Start Guide: Create software robots and automate business processes” , Packt Publishing, 1st Edition 2018.

Online Resources

Sl.No	Website Link
1	https://www.w3schools.com/html/
2	https://www.w3schools.com/xml/
3	https://www.udemy.com/course/robotic-process-automation/
4	https://www.javatpoint.com/rpa
5	https://intellipaat.com/blog/tutorial/rpa-tutorial/
6	https://dev.tutorialspoint.com/uiopath/uiopath_tutorial.pdf